

Name_____

True or false

3 points each

1. (**TRUE**) If $P \neq NP$ then no problem in NP can be solved in polynomial time.
2. (**FALSE**) DFS can be used to find single source shortest path in a graph faster than Dijkstra's algorithm, sometimes.
3. (**TRUE**) In searching a tree, DFS is guaranteed to return a positive (found) or negative (not found) result, whereas BFS is not.
4. (**TRUE**) Given any graph, to find the minimal number of colors to color this graph is unsolvable.
5. (**FALSE**) Any problem can be solved with greedy algorithm can also be solved with dynamic programming.
6. (**TRUE**) Bellman Ford algorithm can only be used on directed graphs.
7. (**TRUE**) Bellman Ford algorithm can be used to detect negative cycles in the graph
8. (**TRUE**) All P problems are also NP problems
9. (**TRUE**) Dijkstra's Algorithm does not allow negative edges
10. (**FALSE**) We can find the longest path in a graph with dynamic programming, but not with greedy algorithm.

Short answer:

1. The following is the characters and their frequency in an article. Find an Huffman Code
5 points

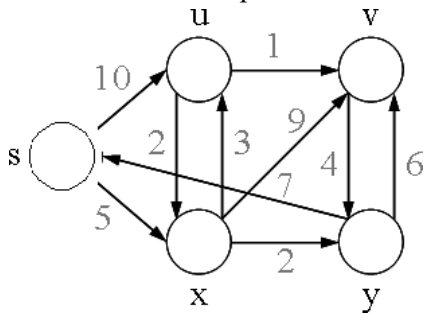
character	Frequency
a	5
b	9
c	8
d	13
e	36
f	45
g	22

Answer:

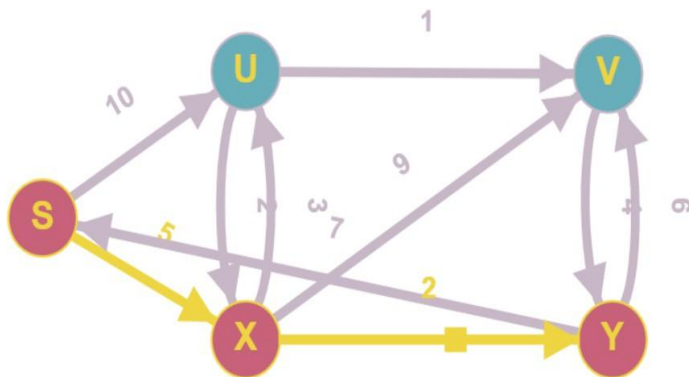
Character	frequency	code
a	5	0100
b	9	0101
c	8	0110
d	13	0111
e	36	10
f	45	11
g	22	00

2. Find the shortest path from S to Y with Dijkstra's algorithm

8 points

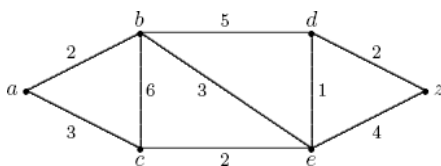


Answer:



The shortest path length is 2: S => X => Y

3. Use Prim's algorithm to find a minimum spanning tree in the following weighted graph. Use alphabetical order to break ties. 5 points



Answer:

S {a} – source

vertices = {b,c,d,e,f}

Step 1: taking lightest edge from source {a,b}

Step 2: after source, s {a,b}, lightest edge = s = {a,b}, {b,e}

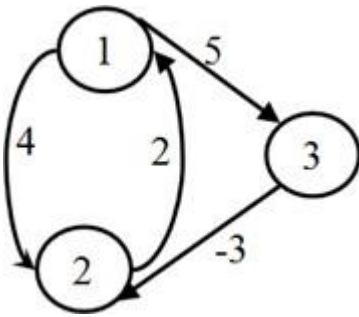
Step 3: Now vertices = {c,d,f}, another edge = {e,d}

Step 4: Point c is left, taking light edge available from e {e,c}

Step 5: vertices = {f}. Taking light edge from d {d,f}

$$\begin{aligned}
 \Rightarrow \text{Total weight} &= \{a,b\} + \{b,e\} + \{c,e\} + \{e,d\} + \{d,f\} \\
 &= 2+3+2+1+2 \\
 &= 10
 \end{aligned}$$

4. For the following graph, run Floyd-Washall algorithm. 10 points



Answer:

4)

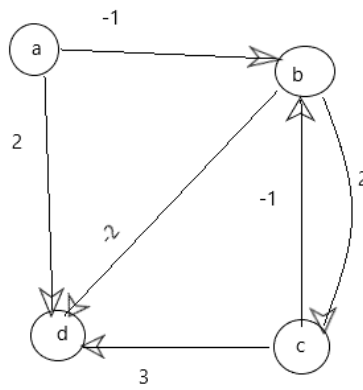
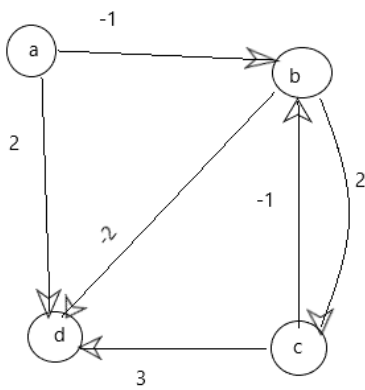
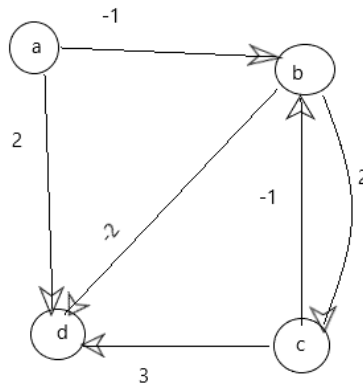
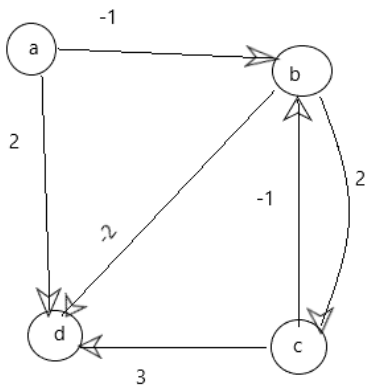
$$A^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & \infty \\ \infty & -3 & 0 \end{bmatrix} \end{matrix}$$

$$A^1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & 7 \\ \infty & 0 & 0 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & 7 \\ \infty & -3 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & 7 \\ -3 & -3 & 0 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & 7 \\ -1 & -3 & 0 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 4 & 5 \\ 2 & 0 & 7 \\ -1 & -3 & 0 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 2 & 5 \\ 2 & 0 & 7 \\ -1 & -3 & 0 \end{bmatrix} \end{matrix}$$

5. Bellman Ford Algorithm. Find shortest path from source a 8 points



Answer:

5) a b c d

 0 -1 1 -3

for from a to:

 b = a → b

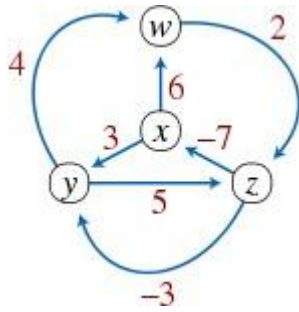
 c = a → b → c

 d = a → b → d

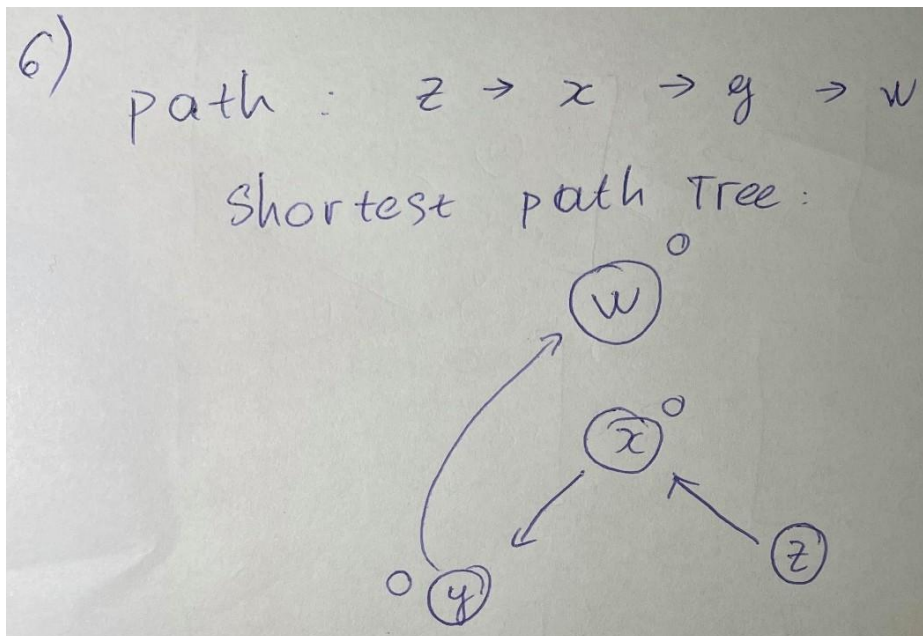
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graph LR
    a((a)) -- -1 --> b((b))
    b((b)) -- 2 --> c((c))
    b((b)) -- -2 --> d((d))
  
```

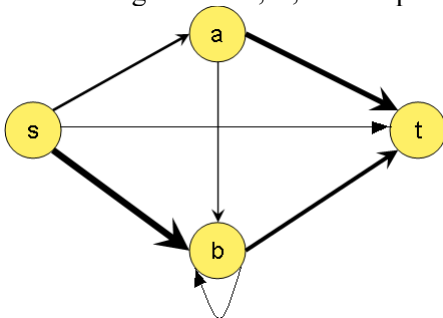
6. Use Johnson's algorithm to find the all source shortest paths in the following graph. 10 points



Answer:

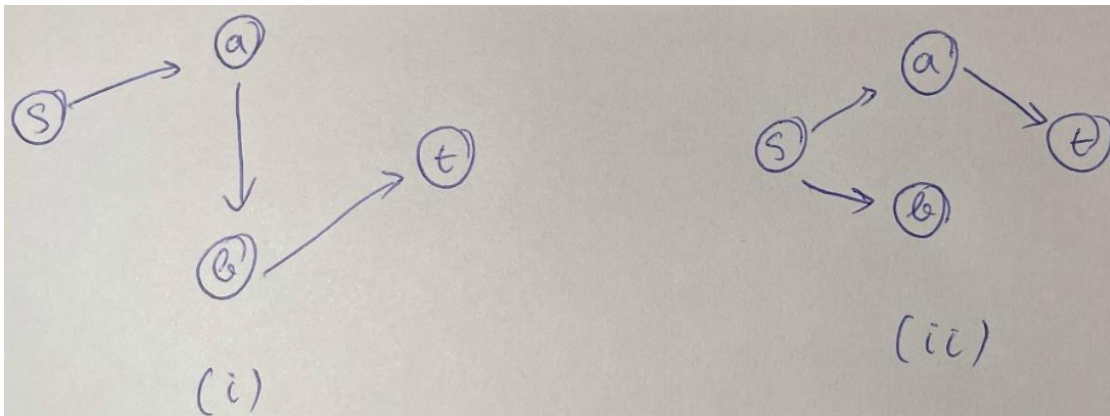


7. Run the DFS on the following graph, with s being the source. Identify the back, cross or forward edges with B, C, or F respectively. 8 points.



Answer:

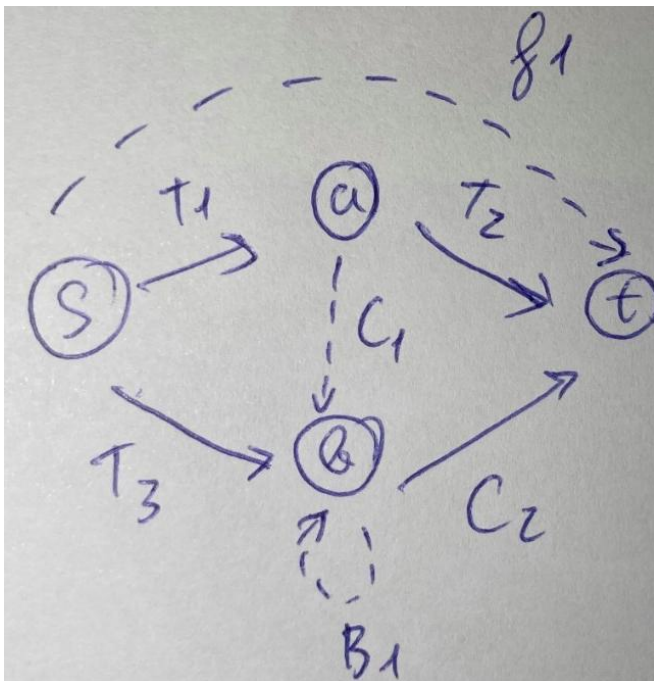
tree edge , backward edge and cross edge we made two DFT of given graph:



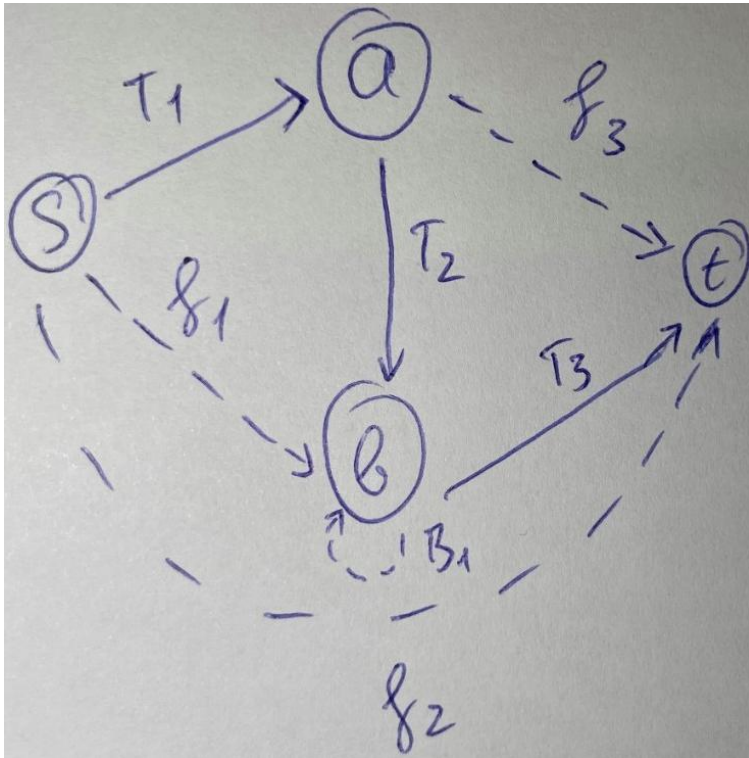
now, in below DFT t_1, t_2, t_3 is showing tree edges.

c_1, c_2, c_3 showing cross edges

f_1, f_2, f_3 is showing forward edges



in above DFT dotted lines are showing tree edges back edges and cross edges.



8. Compute the prefix function for the pattern "ababacab" 8 points

Given Pattern : "ababacab"

We will consider below algorithm for step by step solution to compute prefix function,

Algorithm :

```

p ← "ababacab"
m ← length[p]
A[1] ← 0
k ← 0
for q ← 2 to m do
    while k > 0 and p[k+1] ≠ p[q] do
        k ← A[k]
    end while
    if p[k+1] = p[q] then
        k ← k+1
    end if
    A[q] ← k
end for

```

Stepwise solution for computing prefix function of "ababacab" using above algorithm :

Initially $p \leftarrow \text{"ababacab"}$, $m = \text{length}[p] = 8$, $A[1] = 0$, $k = 0$

Step 1)
q = 2, k = 0
A[2] = 0

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0						

Step 2)
q = 3, k = 0
A[3] = 1

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1					

Step 3)
q = 4, k = 1
A[4] = 2

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1	2				

Step 4)
q = 5, k = 2
A[5] = 3

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1	2	3			

Step 5)
 $q = 6, k = 3$
 $A[6] = 0$

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1	2	3	0		

Step 6)
 $q = 7, k = 0$
 $A[7] = 1$

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1	2	3	0	1	

Step 7)
 $q = 8, k = 1$
 $A[8] = 2$

q	1	2	3	4	5	6	7	8
p	a	b	a	b	a	c	a	b
A	0	0	1	2	3	0	1	2

A represents prefix function, **Answer : Prefix function A :**

0	0	1	2	3	0	1	2
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9. Describe the steps in proving a problem is in NPC

8 points

The reducibility relation " $\leq_T$ " is transitive, i.e.,

$A \leq_T B$ and $B \leq_T C$ imply $A \leq_T C$

Therefore, to prove that a problem A is NPC, we need to:

(1) show that $A \in \text{NP}$

(2) choose NPC problem B , i.e., $B \in \text{NPC}$,

classify a polynomial transformation T from B to A show that $B \leq_T A$